Customer Churn Analysis Report

Overview

Customer churn is often viewed as a customer retention challenge. However, from a business perspective, the more important question is:

What is the business impact of customer churn?

This report analyzes customer churn using the IBM Telco Customer Churn Dataset, focusing on the financial and operational impact of churn rather than simply measuring customer attrition. The analysis aims to identify revenue at risk, understand the drivers of churn, and uncover opportunities for improving customer retention.

Business Questions

The analysis was designed to answer the following business questions:

1. How much revenue is at risk due to churn?
2. Which customers represent the highest future value at risk?
3. Which churn reasons cost the business the most revenue?
4. Which contract types have the highest churn?
5. Which internet services experience the highest churn?
6. Which services are associated with lower churn?
7. Which cities generate the most customer value?

Executive Summary

The analysis identified significant revenue and customer lifetime value losses associated with churn.

Key findings include:

- **$3.69M in revenue has already been lost** due to customer churn.
- **$7.76M in Customer Lifetime Value (CLTV) has been lost**, highlighting the long-term financial impact of churn.
- **Month-to-Month customers account for 88.55% of churned customers**, making them the highest-risk contract segment.
- **Fiber Optic customers generate 81.99% of churned revenue**, representing the most valuable segment experiencing churn.
- The top three churn drivers account for **$1.96M in lost revenue**.
- **446 high risk customers currently contribute $3.02M in revenue**, representing a significant retention opportunity.

Question 1: How Much Revenue Is at Risk Due to Churn?

Segment	Revenue
Churned Customers	$3.69M
Current Customers (Stayed)	$17.65M
New Customers (Joined)	$54K

Analysis

The business has already lost approximately **$3.69 million in revenue** due to churned customers.

From a customer lifetime perspective, churn has resulted in the loss of approximately **$7.76 million in future customer value**, demonstrating that churn affects both current revenue and future profitability.

Business Impact: Reducing churn would have a direct impact on revenue preservation and long term customer value growth.

Question 2: Which Customers Represent the Highest Future Value at Risk?

Metric	Value
High-Risk Customers	446
Revenue Contribution	$3.02M
CLTV	$2.09M
Average Churn Score	72
Average Tenure	58 Months

Analysis

A group of 446 customers has been identified as representing the highest future value at risk.

These customers:

- Generate approximately **14.12% of total revenue**
- Have high predicted churn scores
- Have been customers for nearly five years on average

Business Impact: Targeted retention efforts focused on this segment could protect a significant amount of recurring revenue and future customer lifetime value.

Question 3: Which Churn Reasons Cost the Business the Most Revenue?

Rank	Churn Reason	Lost Revenue
1	Competitor Made Better Offer	$736K
2	Competitor Had Better Devices	$717K
3	Attitude of Support Person	$507K

Analysis

The top three churn reasons account for approximately **$1.96 million in lost revenue**, representing nearly half of all churn-related revenue loss.

Business Impact

The findings suggest that churn is driven by a combination of:

- Competitive pressure
- Product competitiveness
- Customer support experience

Improving value propositions and customer experience may significantly reduce future churn.

Question 4: Which Contract Types Have the Highest Churn?

Distribution of Churned Customers

Contract Type	% of Churned Customers
Month-to-Month	88.55%
One Year	8.88%
Two Year	2.57%

Analysis

Month-to-Month contracts account for nearly nine out of every ten churned customers.

Business Impact: Customers on long term contracts demonstrate substantially higher retention rates. Encouraging customers to migrate toward longer contract commitments may help reduce churn.

Question 5: Which Internet Services Experience the Highest Churn?

Internet Type	% of Churned Customers	% of Churned Revenue
Fiber Optic	66.13%	81.99%
DSL	16.43%	8.48%

Internet Type	% of Churned Customers	% of Churned Revenue
Cable	11.40%	8.36%

Analysis

Fiber Optic customers represent the largest source of churned revenue and account for the majority of churned customers.

Business Impact: Since Fiber Optic customers contribute disproportionately to revenue loss, this segment should be prioritized for churn reduction initiatives.

Question 6: Which Services Are Associated with Lower Churn?

Findings

The following services were most commonly associated with lower churn:

- Online Security
- Premium Tech Support
- Online Backup
- Device Protection Plan

Analysis

Customers subscribed to these services exhibited lower churn rates compared to customers who did not subscribe to them.

Business Impact: Promoting value added services may improve customer engagement and strengthen retention outcomes.

Question 7: Which Cities Generate the Most Customer Value?

City	Revenue Contribution
Los Angeles	~$0.87M
San Diego	~$0.72M
Sacramento	~$0.34M
San Jose	~$0.31M
San Francisco	~$0.30M

Together, Los Angeles and San Diego contribute:

- **15.90% of total revenue**
- **16.33% of total CLTV**

Analysis

These cities represent the highest value geographic markets within the customer base.

Business Impact: Retention strategies focused on these markets are likely to deliver the greatest financial return.

Business Recommendations

Based on the findings, the following actions are recommended:

1. Prioritize Fiber Optic Customer Retention

Fiber Optic customers account for the majority of churned revenue and should be the primary focus of retention initiatives.

2. Reduce Month-to-Month Customer Churn

Introduce incentives that encourage migration toward longer contract terms.

3. Strengthen Competitive Positioning

Address churn driven by competitor offers and device quality through pricing, product, and service improvements.

4. Improve Customer Support Experience

Enhance support processes and training to reduce churn caused by negative service interactions.

5. Promote Value Added Services

Increase adoption of Online Security, Premium Tech Support, Online Backup, and Device Protection services among high risk customers.

Conclusion

This analysis demonstrates that customer churn is not simply a customer retention issue it is a revenue and profitability issue.

By linking churn behavior to revenue, CLTV, customer segments, and churn drivers, the report provides actionable insights that can help the business prioritize retention efforts, protect customer value, and reduce future revenue loss.

The findings highlight the importance of focusing on high value customers, high risk service segments, and the underlying reasons customers choose to leave.